KIRTHIVASAN M R

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PROFESSIONAL SUMMARY

Pre-final year ECE student with a focus on backend development, building projects across multi-agent orchestration systems and RAG pipelines. Placed in two national-level hackathons with hands-on experience using Python, FastAPI, and PostgreSQL. Actively seeking SDE internship or full-time opportunities in backend engineering.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

SRM Institute of Science and Technology | **CGPA: 8.53** **Jul 2023 – Apr 2027**

SKILLS

Programming languages: Python, SQL, HTML, CSS, JavaScript
Backend: FastAPI, Node.js, SQLAlchemy
Databases: PostgreSQL, Firebase, ChromaDB
Libraries: Pandas, Matplotlib, Numpy, LangChain
Tools & Platforms: Git, Docker, AWS, Google Cloud, Linux

WORK EXPERIENCE

Shiksak.com **June 2025 - July 2025**

- Diagnosed and fixed a profile photo upload bug in an Express.js backend, images were saving to local disk with no static serving; rebuilt the storage layer so uploads reflected correctly on the frontend.
- Tested server-side functionality on the VISE platform and resolved critical backend bugs improving stability.

PROJECTS

Orchestrix *(2nd Place — Evolve Hackathon)* Github **Apr 2026**

- Designed a 6-agent orchestration system (FastAPI + SQLAlchemy + PostgreSQL) with intent-based routing; agents independently handle paper discovery, trend analysis, summarization, citation generation, roadmap building, and context-aware Q&A over selected papers.
- Built a multi-source paper discovery pipeline aggregating Semantic Scholar and OpenAlex APIs; fetches up to 100 papers per query with deduplication, ranking, and query expansion, paginated to 20 results per view with full session traces persisted to PostgreSQL.
- Implemented citation export in APA, IEEE, and BibTeX formats and structured PDF session report generation via dedicated export service endpoints.

Invadr (*3rd Place — NXTGEN International Hackathon*) Github **Feb 2026**

- Built an offline-first React Native mobile app with AsyncStorage persistence and NetInfo-triggered background sync; tracked each report across Pending → Syncing → Uploaded states.
- Implemented image capture pipeline with automatic 512×512 resize and JPEG compression to under 300KB, feeding a FastAPI /predict endpoint that returns species name, confidence score, and invasive risk level.
- Designed a Haversine-based geospatial clustering algorithm that triggers High Risk Zone alerts when 5+ high-confidence invasive species reports appear within a 5km radius in 7 days, rendered as overlay circles on a hybrid map.

CERTIFICATIONS

- ‘Certified Python programmer’ by Hacker rank,
- ‘Machine learning with python’ by IBM (coursera)